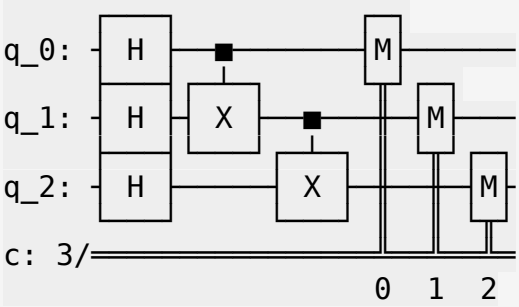
Practical 5

```
from qiskit import QuantumCircuit, transpile
from qiskit_aer import AerSimulator

# Quantum representation of 3-puzzle
qc = QuantumCircuit(3, 3)

# Create superposition
qc.h([0, 1, 2])

# Production rules using CNOT gates
qc.cx(0, 1)
qc.cx(1, 2)

# Measure
qc.measure([0, 1, 2], [0, 1, 2])

simulator = AerSimulator()

compiled = transpile(qc, simulator)

result = simulator.run(compiled, shots=1024).result()

counts = result.get_counts()

print("3-Puzzle Quantum Production System Output:")
print(counts)

print("\nCircuit:")
print(qc.draw())
```

```
3-Puzzle Quantum Production System Output:
{'110': 133, '001': 119, '101': 116, '111': 128, '011': 135, '010':
135, '000': 127, '100': 131}

Circuit:
```